Assessment 2

DDD 2: Introduction to Programming and Basic Data Analysis

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2026-11-03

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Setup

```
rm(list = ls())
```

Load Required Libraries

```
library(tibble)
library(dplyr)
library(rlang)
library(lobstr)
library(rio)
library(ggplot2)
```

```
library(labelled)
library(stargazer)
library(scales)
library(ivo.table)
library(flextable)
library(janitor)
library(tinytex)
```

Exam Environment

```
ASSESSMENT2_DDD <- new.env()
```

Question 1

```
ASSESSMENT2_DDD <- new.env()

ASSESSMENT2_DDD$x <- 5

while(ASSESSMENT2_DDD$x <= 15) {
  print(ASSESSMENT2_DDD$x)
  ASSESSMENT2_DDD$x <- ASSESSMENT2_DDD$x + 2
}
```

```
[1] 5
[1] 7
[1] 9
[1] 11
[1] 13
[1] 15
```

Question 2 (a) Explanitation of the code

```
x<- 13
repeat{
  x= x+2
  print(x)
  if ( x == 25 ) break
}
```

```
[1] 15
[1] 17
[1] 19
[1] 21
[1] 23
[1] 25
```

```
# During the execution of the code variable x starts with a value of 13. In each iteration of the repeat loop, x is incremented by 2 and its new value is printed. The loop continues until x reaches 25, at which point the break statement is executed, terminating or we can say stoping the loop. The output will be like as printed in the console:
```

Question 2 (b) Modification of the code

```
x <- 13
repeat{
  print(x)
  x = x + 2
  if (x > 25) break
}
```

```
[1] 13
[1] 15
[1] 17
[1] 19
[1] 21
[1] 23
[1] 25
```

Question 3 (a) Creating a data frame with 2 columns and 6 rows

```
df <- tibble::tibble(
  a = rnorm(6),
  b = rnorm(6))

df
```

```
# A tibble: 6 × 2
       a       b
```

```
   <dbl>   <dbl>
1  0.766 -0.295
2 -1.10  -0.641
3  1.26   0.468
4 -0.235 -0.982
5 -1.29  -0.0573
6  0.609 -0.956
```

Question 3 (b) function defining to calculate mean deviation

```
mean_dev <- function(x) {
  x - mean(x, na.rm = TRUE)}
mean_dev(df$a)
```

```
[1]  0.7639419 -1.1038446  1.2584444 -0.2375372 -1.2883527  0.6073482
```

```
mean_dev(df$b)
```

```
[1]  0.1155317 -0.2304310  0.8786796 -0.5716363  0.3530831 -0.5452272
```

```
## The output of the above code will be the mean deviation of each value in columns a and b from their means. The mean deviation is calculated by subtracting the mean of the column from each individual value in that column. The output will be displayed here;
```

Question 3 (c)

```
library(purrr)

df$c <- mean_dev(df$a)
df$d <- mean_dev(df$b)

df
```

```
# A tibble: 6 × 4
       a       b      c      d
   <dbl>   <dbl>  <dbl>  <dbl>
1  0.766 -0.295   0.764  0.116
2 -1.10  -0.641  -1.10  -0.230
```

```
3  1.26   0.468   1.26   0.879
4 -0.235 -0.982  -0.238 -0.572
5 -1.29  -0.0573 -1.29   0.353
6  0.609 -0.956   0.607 -0.545
```

Question 4 (IHDS Data exercise)

```
library(tidyverse)
library(rio)
HH_DATA <- import("../data/ddd2/RAW_DATA/ICPSR_36151/HH_DATA/HH_DATA.dta")

IHDS<- subset(HH_DATA, select= c(ID18C, ID13, URBAN2011))
view(IHDS)
```

Question 4 (Data Cleaning and Variable Labeling)

```
library(labelled)

unique(IHDS$ID18C)
```

```
 [1]  0 10  4  8 16  2  9  5 NA  3 12  6  7 11 15  1 14 13
```

```
table(IHDS$ID18C)
```

```

    0     1     2     3     4     5     6     7     8     9    10    11    12
25569   214  1502  1271  1817  3712   586  1050  1606   567  2206   149   697
   13    14    15    16
   27    22   565   251
```

```
attributes(IHDS$ID18C)
```

```
NULL
```

```
unique(IHDS$ID13)
```

```
[1]  3 NA  2  5  6  4  1
```

```
table(IHDS$ID13)
```

```
    1     2     3     4     5     6
 2192  9665 17056  8941  3644   568
```

```
attributes(IHDS$ID13)
```

```
NULL
```

```
unique(IHDS$URBAN2011)
```

```
[1] 0 1
```

```
table(IHDS$URBAN2011)
```

```
    0     1
27579 14573
```

```
attributes(IHDS$URBAN2011)
```

```
NULL
```

```
IHDS <- IHDS |> mutate(ID18C = ifelse(ID18C == 16, NA, ID18C))

IHDS <- IHDS |> filter(ID18C <= 15 & ID18C >= 0 | is.na(ID18C))

var_label(IHDS$ID18C) <- "Years of Education"
var_label(IHDS$ID13) <- "Caste Category"
var_label(IHDS$URBAN2011) <- "Urban Residence"

IHDS$ID18C <- as.numeric(IHDS$ID18C)
IHDS$ID13 <- as.factor(IHDS$ID13)
IHDS$URBAN2011 <- as.factor(IHDS$URBAN2011)
```

```
table(IHDS$ID18C)
```

```
    0     1     2     3     4     5     6     7     8     9    10    11    12
25569   214  1502  1271  1817  3712   586  1050  1606   567  2206   149   697
   13    14    15
   27    22   565
```

```
table(IHDS$ID13)
```

```
    1     2     3     4     5     6
 2192  9665 17056  8941  3644   568
```

```
table(IHDS$URBAN2011)
```

```
    0     1
27579 14573
```

```
view(IHDS)
```

Question 4 (Recoding Caste Variable)

```
unique(IHDS$ID13)
```

```
[1] 3    <NA> 2    5    6    4    1
Levels: 1 2 3 4 5 6
```

```
table(IHDS$ID13)
```

```
    1     2     3     4     5     6
 2192  9665 17056  8941  3644   568
```

```
IHDS <- IHDS |>
  mutate(caste_r = case_when(
    ID13 == 1 | ID13 == 2 ~ 1,
    ID13 == 3 ~ 3,
    ID13 == 4 ~ 4,
    ID13 == 5 ~ 5,
    TRUE ~ NA_real_
  ))

table(IHDS$caste_r)
```

```
    1     3     4     5
11857 17056  8941  3644
```

```
var_label(IHDS$caste_r) <- "Caste Category Recoded"
val_labels(IHDS$caste_r) <- c(
  General = 1,
  OBC = 3,
  SC = 4,
  ST = 5)

attributes(IHDS$caste_r)
```

```
$labels
General     OBC      SC      ST
      1       3       4       5

$label
[1] "Caste Category Recoded"

$class
[1] "haven_labelled" "vctrs_vctr"     "double"
```

Question 4 (SUmmary Statistics)

```
library(stargazer)

IHDS_summary <- IHDS |>
  select(ID18C, caste_r, URBAN2011) |>
```

```
  mutate(
    ID18C = as.numeric(ID18C),
    caste_recoded = as.numeric(caste_r),
    URBAN2011 = as.numeric(URBAN2011)
  )

IHDS_summary |>
  as.data.frame() |>
  stargazer(
    type = "text",
    title = "Summary Statistics",
    summary.stat = c("n", "mean", "sd", "min", "max"),
    digits = 2,
    out = "Results/summary_statistics.txt"
  )
```

```
Summary Statistics
==========================================
Statistic       N    Mean St. Dev. Min Max
------------------------------------------
ID18C         41,560 2.48   3.74    0  15
caste_r       41,498 2.82   1.30    1   5
URBAN2011     42,152 1.35   0.48    1   2
caste_recoded 41,498 2.82   1.30    1   5
------------------------------------------
```

Extra code (not run)

```
# STEP 1: Create family size and constructed per capita income
FINAL_EXAM$HH_DATA_q1 <- HH_DATA |>
  mutate(
     family_size = NADULTM + NADULTF + NCHILDM + NCHILDF + NTEENM + NTEENF +
NELDERM + NELDERF,
    income_pc_constructed = INCOME / family_size
  ) |>
  select(INCOME, family_size, income_pc_constructed, INCOMEPC)

# Check for missing values
summary(FINAL_EXAM$HH_DATA_q1)
```

```r
# STEP 2: Remove rows with missing or zero values
FINAL_EXAM$df_q1_clean <- FINAL_EXAM$HH_DATA_q1 |>
  filter(!is.na(income_pc_constructed),
         !is.na(INCOMEPC),
         income_pc_constructed > 0,
         INCOMEPC > 0)

# STEP 3: Plot correlation between constructed and given per capita income
FINAL_EXAM$df_q1_clean |>
  ggplot(aes(x = INCOMEPC, y = income_pc_constructed)) +
  geom_point(alpha = 0.5, size = 2) +
  geom_smooth(se = TRUE, method = "lm", color = "red") +
  labs(
    title = "Correlation between Constructed and Given Per Capita Income (Ramesh
Beniwal)",
    x = "Given Per Capita Income (Rupees) [Ramesh Beniwal]",
    y = "Constructed Per Capita Income (Rupees) [Ramesh Beniwal]"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold", size = 12),
    axis.title = element_text(size = 10)
  )

# STEP 4: Calculate correlation coefficient
FINAL_EXAM$correlation_q1 <- cor(FINAL_EXAM$df_q1_clean$INCOMEPC,
                                 FINAL_EXAM$df_q1_clean$income_pc_constructed)
print(paste("Correlation coefficient:", round(FINAL_EXAM$correlation_q1, 4)))
```

```r
# ====================================================
# QUESTION 2: Validate Income-Consumption Assumption
# ====================================================

# STEP 1: Prepare data with both income variables
FINAL_EXAM$HH_DATA_q2 <- HH_DATA |>
  mutate(
     family_size = NADULTM + NADULTF + NCHILDM + NCHILDF + NTEENM + NTEENF +
NELDERM + NELDERF,
    income_pc_constructed = INCOME / family_size
  ) |>
  select(income_pc_constructed, INCOMEPC, COPC) |>
  filter(income_pc_constructed > 0, INCOMEPC > 0, COPC > 0)

# Check data
print("Summary before outlier removal (Q2):")
summary(FINAL_EXAM$HH_DATA_q2)

# STEP 2: Remove outliers (filter high values like you did in scripts)
FINAL_EXAM$HH_DATA_q2 <- FINAL_EXAM$HH_DATA_q2 |>
  filter(COPC < 100000, income_pc_constructed < 100000, INCOMEPC < 100000)

# Check cleaned data
print("Summary after outlier removal (Q2):")
summary(FINAL_EXAM$HH_DATA_q2)

# STEP 3: PLOT 1 - Both relationships on same graph
FINAL_EXAM$HH_DATA_q2 |>
  ggplot() +
  geom_smooth(aes(x = INCOMEPC, y = COPC, color = "Given Per Capita Income"),
              se = TRUE, method = "lm", linewidth = 1.2) +
  geom_smooth(aes(x = income_pc_constructed, y = COPC, color = "Constructed Per
Capita Income"),
              se = TRUE, method = "lm", linewidth = 1.2) +
  scale_x_continuous(labels = scales::comma) +
  scale_y_continuous(labels = scales::comma) +
  labs(
    title = "Income-Consumption Relationship: Validation of Assumption (Ramesh
Beniwal)",
```

```
    x = "Per Capita Income (Rupees) [Ramesh Beniwal]",
    y = "Per Capita Consumption (Rupees) [Ramesh Beniwal]",
    color = "Income Measure [Ramesh Beniwal]"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold", size = 12),
    axis.title = element_text(size = 10),
    legend.position = "bottom"
  )

# STEP 4: PLOT 2 - Faceted comparison
FINAL_EXAM$HH_DATA_q2_long <- FINAL_EXAM$HH_DATA_q2 |>
  pivot_longer(cols = c(INCOMEPC, income_pc_constructed),
               names_to = "income_type",
               values_to = "income_value") |>
  mutate(income_type = case_when(
    income_type == "INCOMEPC" ~ "Given Per Capita Income",
    income_type == "income_pc_constructed" ~ "Constructed Per Capita Income"
  ))

FINAL_EXAM$HH_DATA_q2_long |>
  ggplot(aes(x = income_value, y = COPC, color = income_type)) +
  geom_point(alpha = 0.3, size = 1.5) +
  geom_smooth(se = TRUE, method = "lm", linewidth = 1) +
  facet_wrap(~income_type, scales = "free_x") +
  scale_x_continuous(labels = scales::comma) +
  scale_y_continuous(labels = scales::comma) +
  labs(
    title = "Income-Consumption Relationship by Income Type (Ramesh Beniwal)",
    x = "Per Capita Income (Rupees) [Ramesh Beniwal]",
    y = "Per Capita Consumption (Rupees) [Ramesh Beniwal]",
    color = "Income Measure [Ramesh Beniwal]"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold", size = 12),
    axis.title = element_text(size = 10),
    legend.position = "bottom"
  )

# STEP 5: Calculate correlations
FINAL_EXAM$cor_given_income       <-       cor(FINAL_EXAM$HH_DATA_q2$INCOMEPC,
```

```
FINAL_EXAM$HH_DATA_q2$COPC)
FINAL_EXAM$cor_constructed_income                                    <-
cor(FINAL_EXAM$HH_DATA_q2$income_pc_constructed, FINAL_EXAM$HH_DATA_q2$COPC)

# STEP 6: Print results
print("QUESTION 2: VALIDATION OF INCOME-CONSUMPTION ASSUMPTION")
print(paste("Correlation (Given Per Capita Income vs Consumption):",
round(FINAL_EXAM$cor_given_income, 4)))
print(paste("Correlation (Constructed Per Capita Income vs Consumption):",
round(FINAL_EXAM$cor_constructed_income, 4)))
print("INTERPRETATION: ASSUMPTION VALIDATED - Both income measures show POSITIVE
correlation with consumption")
print("As per capita income increases, per capita consumption also increases")
```